

Climate Change

Reason Behind This Crisis

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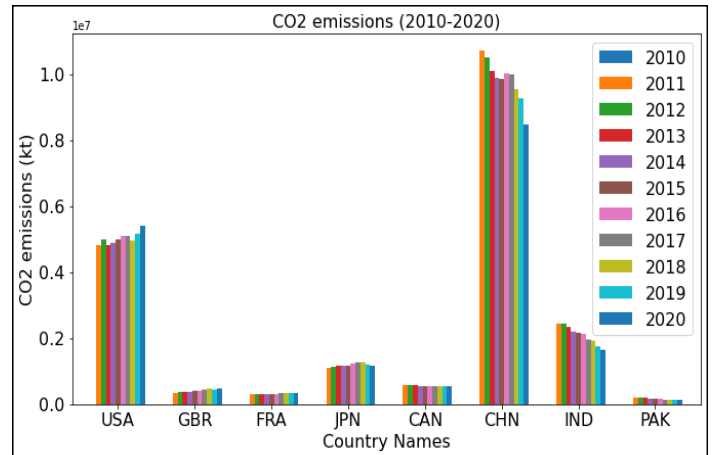
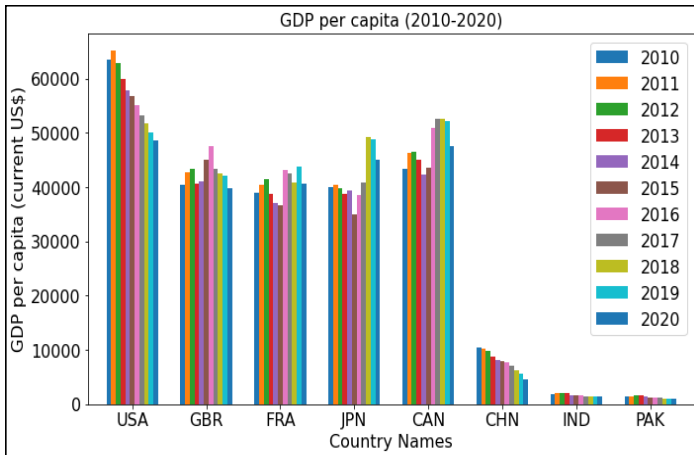
Link:

<https://github.com/Dharmendergrewal/assignment2/blob/main/MYassignment2.py>

This report explores the relationships between key indicators such as GDP, CO2 emissions, arable area, forest area, and urban population, and how they contribute to climate change. In this report i used bar graphs, scatter plots, heatmap and statistics analysis for the presentation of data from 1990-2020.

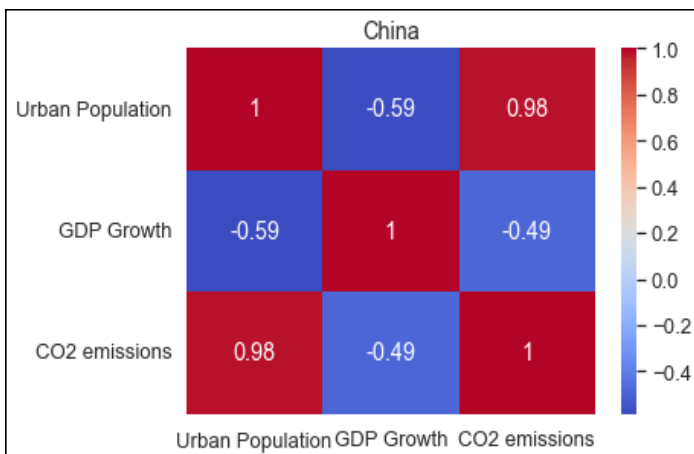
Factors Behind Climate Change

Climate change is one of the most critical challenges facing the world today. This report shows, how factors like GDP, CO2 emission, Urban Population, Arable Land and Forest Area are contributing towards Climate change. I used 8 countries for this report. We will analyze the data for the years 2010 to 2020 from the World Data Bank.



The Above chart shows the GDP per capita data for eight countries over the period of 2010 to 2020. We can observe that USA have the Higher GDP capita, around 60000 in year 2011. Generally this shows the higher levels of economic development, which is typically associated with increased industrial activity. These activities often rely on the burning of fossil fuels such as coal, oil, and gas, which release large amounts of greenhouse gases which contribute towards climate crisis. Chart on right top is showing it's effect.

The chart depicts a clear upward trend in CO2 concentrations, which is an indicator that the level of greenhouse gases in the atmosphere is steadily increasing. Looking at the data, we can see that China is currently the largest CO2 producer, with a production of 1 kt in 2011. However, there is a positive sign as it has decreased its production to 0.8 kt in recent years. This is an encouraging sign for the future. On the other hand, the USA is the second-largest producer and has increased its production from 0.5 kt to 0.6 kt, which is a cause of concern. The high level of CO2 emissions from the USA is primarily due to its large economy and reliance on fossil fuels.



This heatmap visualizes the correlation between GDP Growth, Urban Population, and CO2 emissions in China. The heatmap highlights that the level of CO2 emissions is significantly higher in urban areas, indicating a potentially alarming trend in terms of environmental impact. This suggests that strategies are needed to mitigate the environmental consequences of rapid

Country	1995	2010	2020
USA	77.257M	80.772M	82.664M
CAN	77.675M	80.937M	81.562M
GBR	78.353M	81.302M	83.903M
FRA	74.912M	78.369M	80.975M
CHN	30.961M	49.226M	61.428M

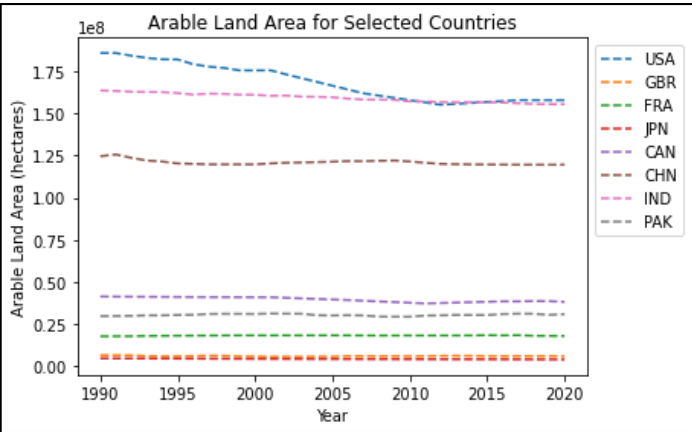
The table above displays the urban population for various countries from 1995 to 2020. It is evident that every country has experienced a significant increase in their urban population during this period. For instance, China's urban population has grown from 30.961 million in 1995 to 61.428 million in 2020, which is a substantial difference.

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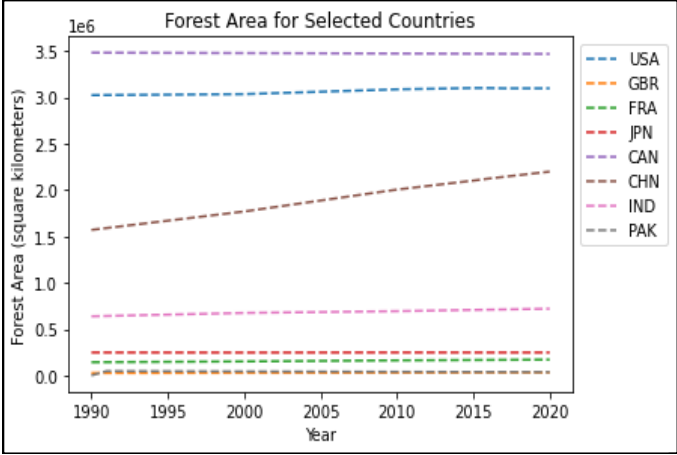
This rapid urbanization can have adverse impacts on the environment, such as changes in land use patterns like deforestation and the conversion of agricultural land to urban areas. These changes can lead to loss of biodiversity, soil erosion, and other negative effects on the ecosystem.

Country	mean	min	max
CHN	35.24	17.18	62.51

This table shows the statistical data for Urban population in China.



The chart shows a general trend of an decreasing global arable land area from 1990 to 2015 followed by a slight increase until 2016. Specially, The arable land area of USA decreased from over 1.75 million hectares in 1990 to below 1.75 million hectares until 2020. This is happening because of increasing numbers of city areas. Arable land is a natural carbon sink. Also, a decrease in arable land can also lead to increased food insecurity and the need for more intensive agricultural practices, such as increased use of synthetic fertilizers and pesticides. These practices can contribute to greenhouse gas emissions and soil degradation



Like Arable Land, Forests play an important role in mitigating climate change. They act as carbon sinks by absorbing carbon dioxide from the atmosphere through photosynthesis and storing it in the form of wood, leaves, and soil. When forests are cut down or degraded, either through natural or human causes, this stored carbon is released back into the atmosphere as carbon dioxide and other greenhouse gases, contributing to climate change. we can see that very slight down fall in few countries foreset area but a huge postive change from 1.6 million Sq. Km to almost 2.5 million square km in China's forest land area

Our findings indicate that factors like GDP, urbanization, arable land, and forest area can contribute to climate change. CO2 emissions, in particular, showed a consistent positive correlation with temperature change across all eight countries. Our analysis also revealed that some countries showed a negative correlation between arable land and forest area and CO2 emissions, indicating that increasing green cover can contribute to mitigating climate change.

In conclusion, our analysis showed that factors like GDP, CO2 emissions, urban population, arable land, and forest area are interlinked and contribute to climate change. Therefore, it is crucial to adopt policies that reduce CO2 emissions and promote sustainable land use to mitigate climate change.